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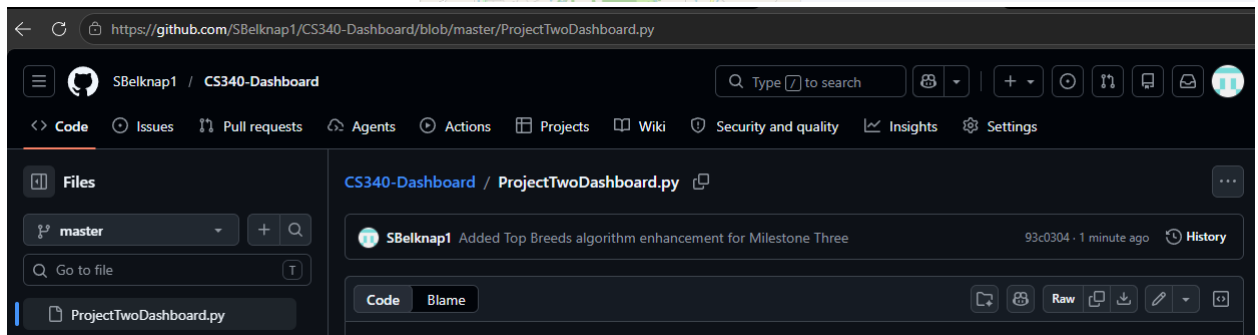
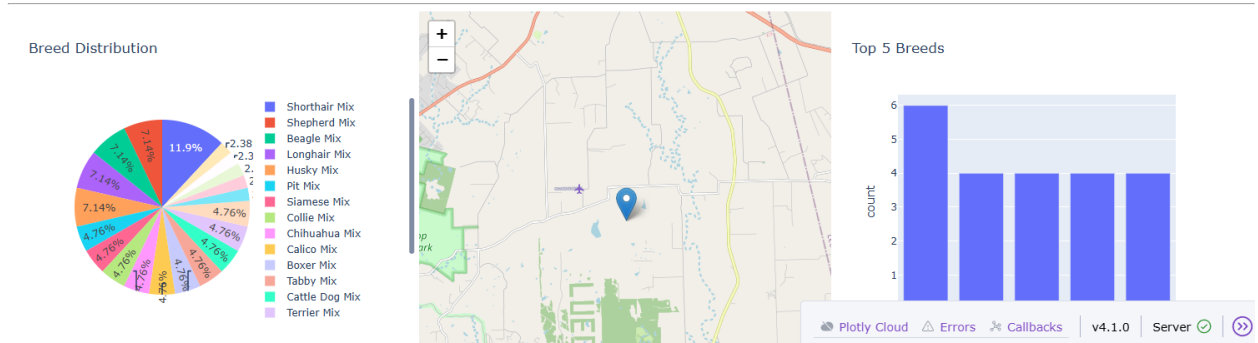
CS-499

Milestone Three – Algorithms and Data Structures

My artifact for this milestone is my CS-340 dashboard project, originally created in the Programming Concepts course. The dashboard uses Python, Dash, and MongoDB to process and visualize animal shelter data. It includes a data table, a pie chart, and an interactive map. This artifact contains several opportunities to demonstrate algorithmic thinking and data-structure manipulation, which is why I selected it for the Algorithms and Data Structures enhancement.

I included this artifact in my ePortfolio because it showcases my ability to design and implement data-driven features using algorithmic principles. For this milestone, I enhanced the artifact by adding a new “Top Breeds” algorithm. This algorithm reads the dataset, counts how many times each breed appears using a dictionary (hash map), sorts the results in descending order, extracts the top five breeds, and displays them in a bar chart. This enhancement demonstrates my ability to work with data structures, implement counting and sorting algorithms, and integrate algorithmic logic into a real application.

age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat
filter data...												
1 year	A123456	Dog	Labrador Retriever Mix	Black	2014-01-01	2015-01-01 12:34:56	2015-01-01T12:34:56	Buddy		Adoption	Neutered Male	30.12345
2 years	A765432	Cat	Domestic Shorthair Mix	Brown Tabby	2018-05-12	2020-06-15 14:22:00	2020-06-15T14:22:00	Whiskers		Adoption	Spayed Female	30.2672
2 years	A000001	Dog	Labrador Mix	Black	2018-01-10	2020-02-15 10:22	2020-02-15T10:22	Shadow		Adoption	Neutered Male	30.2672
1 year	A000002	Cat	Shorthair Mix	Brown	2019-03-12	2020-04-20 14:10	2020-04-20T14:10	Mittens		Transfer	Spayed Female	30.3001
3 years	A000003	Dog	Shepherd Mix	Tan	2017-06-01	2020-07-10 09:45	2020-07-10T09:45	Rex	Foster	Return to Owner	Intact Male	30.25
4 months	A000004	Dog	Pit Mix	White	2020-01-01	2020-05-01 12:00	2020-05-01T12:00	Spot		Adoption	Neutered Male	30.28
6 years	A000005	Cat	Siamese Mix	Cream	2014-08-15	2020-09-10 11:30	2020-09-10T11:30	Luna		Euthanasia	Spayed Female	30.26
8 months	A000006	Dog	Beagle Mix	Tri	2019-11-01	2020-07-01 08:00	2020-07-01T08:00	Scout		Adoption	Intact Male	30.29
5 years	A000007	Dog	Collie Mix	Black/White	2015-05-05	2020-10-12 15:00	2020-10-12T15:00	Dash	Foster	Transfer	Neutered Male	30.31
10 months	A000008	Cat	Longhair Mix	Gray	2019-09-01	2020-07-20 13:00	2020-07-20T13:00	Smokey		Adoption	Intact Female	30.27



This enhancement aligns with the Computer Science program outcome related to designing and evaluating computing solutions using algorithmic principles. By implementing a custom algorithm and integrating it into the dashboard’s callback system, I demonstrated my ability to transform data, optimize logic, and manage trade-offs in design choices. This enhancement fulfills the outcome I planned to meet in Module One.

Enhancing this artifact taught me how algorithmic logic can significantly improve the usefulness and clarity of a data-driven application. One challenge I faced was ensuring that the algorithm integrated smoothly with the existing Dash callbacks without causing performance issues or errors when the dataset was empty. I also had to consider edge cases, such as missing breed values or filters that returned no results. Through this process, I strengthened my understanding of algorithm design, data transformation, and defensive programming. Overall, this enhancement improved both the functionality of the dashboard and my skills in algorithms and data structures.

